

Technological unemployment

Technological unemployment is the loss of jobs due to technological change.^{[1][2][3]} It is a key type of structural unemployment. Technological change typically includes the introduction of labour-saving "mechanical-muscle" machines or more efficient "mechanical-mind" processes (automation) and, in doing so, humans' role in these processes are minimized.^[4] Historical examples include artisan weavers losing work due to the introduction of mechanized looms, leading to protests by the Luddites. A contemporary example of technological unemployment is the displacement of retail cashiers by self-service tills and cashierless stores.

Technological change can cause short-term job losses in a specific profession or industry. Whether it leads to lasting increases in unemployment has been debated. The phrase "technological unemployment" was popularised by John Maynard Keynes in the 1930s, who said it was "only a temporary phase of maladjustment".^[5]

Advances in artificial intelligence (AI) have reignited debates about the possibility of mass unemployment, or even the end of employment altogether. Some experts, such as Geoffrey Hinton, believe that the development of artificial general intelligence and advanced robotics will eventually enable the automation of all intellectual and physical tasks, suggesting the need for a basic income for non-workers to subsist.^{[6][7]} Others, like Daron Acemoglu, argue that humans will remain necessary for certain tasks, or complementary to AI, disrupting the labor market without necessarily causing mass unemployment.^{[8][9]} The World Bank's 2019 World Development Report argues that while automation displaces workers, technological innovation creates more new industries and jobs on balance.^[10]



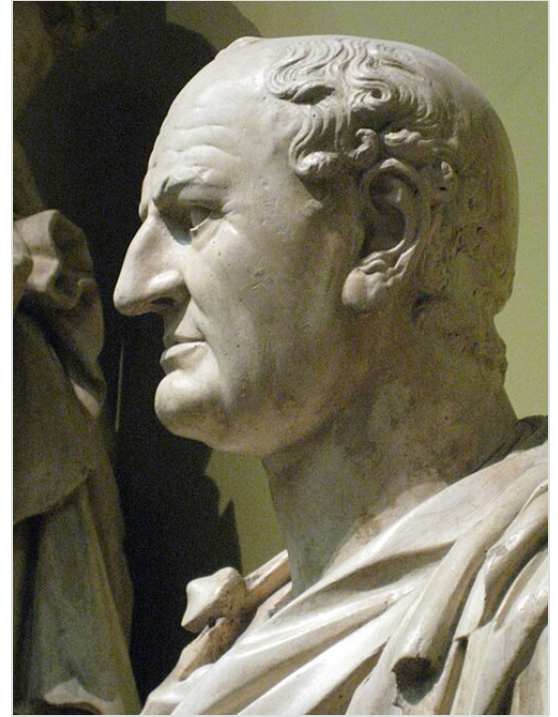
A pharmacy robot delivering to a nurses' station at San Jose's Good Samaritan Hospital in the United States in 2008: In the 21st century, robots are beginning to perform roles not just in manufacturing but also in the service sector – in healthcare, for example.

History

Classical era

The issue of machines displacing human labour has been discussed since at least Aristotle's time.^{[11][12]}

According to author Gregory Woirol, the phenomenon of technological unemployment is likely to have existed since at least the invention of the wheel.^[13] Ancient societies had various methods for relieving the poverty of those unable to support themselves with their own labour. Ancient China and ancient Egypt may have had various centrally run relief programmes in response to technological unemployment dating back to at least the second millennium BC.^[14] Ancient Hebrews and adherents of the ancient Vedic religion had decentralised responses where aiding the poor was encouraged by their faiths.^[14] In ancient Greece, free labourers could find themselves unemployed due to both the effects of ancient labour saving technology and to competition from slaves ("machines of flesh and blood"^[15]). Sometimes, these unemployed workers would starve to death or were forced into slavery themselves although in other cases they were supported by handouts. Pericles responded to perceived technological unemployment by launching public works programmes to provide paid work to the jobless. Pericle's programmes were criticized for wasting public money but these criticisms were defeated.^[16]



Roman Emperor Vespasian, who refused a low-cost method of transport of heavy goods that would put laborers out of work

Perhaps the earliest example of a scholar discussing the phenomenon of technological unemployment occurs with Aristotle, who speculated in Book One of *Politics* that if machines could become sufficiently advanced, there would be no more need for human labour.^[17] Similar to the Greeks, ancient Romans responded to the problem of technological unemployment by relieving poverty with handouts (such as the *Cura Annonae*). Several hundred thousand families were sometimes supported like this at once.^[14] Less often, jobs were directly created with public works programmes, such as those launched by the Gracchi. Various emperors even went as far as to refuse or ban labour saving innovations.^{[18][19]} In one instance, the introduction of a labor-saving invention was blocked, when Emperor Vespasian refused to allow a new method of low-cost transportation of heavy goods, saying "You must allow my poor hauliers to earn their bread."^[20] Labour shortages began to develop in the Roman empire towards the end of the second century AD, and from this point mass unemployment in Europe appears to have largely receded for over a millennium.^[21]

Post-classical era

The medieval and early renaissance period saw the widespread adoption of newly invented technologies, as well as older ones which had been conceived yet barely used in the Classical era.^[22] Some were invented in Europe while others were invented in more Eastern countries like China, India, Arabia and Persia. The Black Death left fewer workers across Europe. Mass unemployment began to reappear in Europe, especially in Western, Central and Southern Europe in the 15th century, partly as a result of population growth, and partly due to changes in the availability of land for subsistence farming caused by early enclosures.^[23] As a result of the threat of unemployment, there was less tolerance for disruptive new technologies. European authorities would often side with groups representing subsections of the working population, such as Guilds, banning new technologies and sometimes even executing those who tried to promote or trade in them.^[24]

16th to 18th century

In Great Britain, the ruling elite began to take a less restrictive approach to innovation somewhat earlier than in much of continental Europe, which has been cited as a possible reason for Britain's early lead in driving the Industrial Revolution.^[25] Yet concern over the impact of innovation on employment remained strong through the 16th and early 17th century. A famous example of new technology being refused occurred when the inventor William Lee invited Queen Elizabeth I to view a labour saving knitting machine. The Queen declined to issue a patent on the grounds that the technology might cause unemployment among textile workers. After moving to France and also failing to achieve success in promoting his invention, Lee returned to England but was again refused by Elizabeth's successor James I for the same reason.^[26]

After the Glorious Revolution, authorities became less sympathetic to workers concerns about losing their jobs due to innovation. An increasingly influential strand of Mercantilist thought held that introducing labour saving technology would actually reduce unemployment, as it would allow British firms to increase their market share against foreign competition. From the early 18th century workers could no longer rely on support from the authorities against the perceived threat of technological unemployment. They would sometimes



Elizabeth I, who refused to patent a knitting machine invented by William Lee, saying "Consider thou what the invention could do to my poor subjects. It would assuredly bring them to ruin by depriving them of employment, thus making them beggars."

take direct action, such as machine breaking, in attempts to protect themselves from disruptive innovation. Joseph Schumpeter notes that as the 18th century progressed, thinkers would raise the alarm about technological unemployment with increasing frequency, with von Justi being a prominent example.^[27] Yet Schumpeter also notes that the prevailing view among the elite solidified on the position that technological unemployment would not be a long-term problem.^[26]^[23]

19th century

It was only in the 19th century that debates over technological unemployment became intense, especially in Great Britain where many economic thinkers of the time were concentrated. Building on the work of Dean Tucker and Adam Smith, political economists began to create what would become the modern discipline of economics.^[note 1] While rejecting much of mercantilism, members of the new discipline largely agreed that technological unemployment would not be an enduring problem. In the first few decades of the 19th century, several prominent political economists did, however, argue against the optimistic view, claiming that innovation could cause long-term unemployment. These included Sismondi,^[28] Malthus, J S Mill, and from 1821, David Ricardo himself.^[29] As arguably the most respected political economist of his age, Ricardo's view was challenging to others in the discipline. The first major economist to respond was Jean-Baptiste Say, who argued that no one would introduce machinery if they were going to reduce the amount of product,^[note 2] and that as Say's law states that supply creates its own demand, any displaced workers would automatically find work elsewhere once the market had had time to adjust.^[30] Ramsey McCulloch expanded and formalised Say's optimistic views on technological unemployment, and was supported by others such as Charles Babbage, Nassau Senior and many other lesser known political economists. Towards the middle of the 19th century, Karl Marx joined the debates. Building on the work of Ricardo and Mill, Marx went much further, presenting a deeply pessimistic view of technological unemployment; his views attracted many followers and founded an enduring school of thought but mainstream economics was not dramatically changed. By the 1870s, at least in Great Britain, technological unemployment faded both as a popular concern and as an issue for academic debate. It had become increasingly apparent that innovation was increasing prosperity for all sections of British society, including the working class. As the classical school of thought gave way to neoclassical economics, mainstream thinking was tightened to take into account and refute the pessimistic arguments of Mill and Ricardo.^[31]

20th century

For the first two decades of the 20th century, mass unemployment was not the major problem it had been in the first half of the 19th. While the Marxist school and a few other thinkers continued to challenge the optimistic view, technological unemployment was not a significant concern for mainstream economic thinking until the mid to late 1920s. In the 1920s mass unemployment re-

emerged as a pressing issue within Europe. At this time the U.S. was generally more prosperous, but even there urban unemployment had begun to increase from 1927. Rural American workers had been suffering job losses from the start of the 1920s; many had been displaced by improved agricultural technology, such as the tractor. The centre of gravity for economic debates had by this time moved from Great Britain to the United States, and it was here that the 20th century's two great periods of debate over technological unemployment largely occurred.^[32]



The peak periods for the two debates were in the 1930s and the 1960s. According to economic historian Gregory R Woirol, the two episodes share several similarities.^[33] In both cases academic debates were preceded by an outbreak of popular concern, sparked by recent rises in unemployment. In both cases the debates were not conclusively settled, but faded away as unemployment was reduced by an outbreak of war – World War II for the debate of the 1930s, and the Vietnam War for the 1960s episodes. In both cases, the debates were conducted within the prevailing paradigm at the time, with little reference to earlier thought. In the 1930s, optimists based their arguments largely on neo-classical beliefs in the self-correcting power of markets to reduce any short-term unemployment via compensation effects. In the 1960s, belief in compensation effects was less strong, but the mainstream Keynesian economists of the time largely believed government intervention would be able to counter any persistent technological unemployment that was not cleared by market forces. Another similarity was the publication of a major Federal study towards the end of each episode, which broadly found that long-term technological unemployment was not occurring (though the studies did agree innovation was a major factor in the short term displacement of workers, and advised government action to provide assistance).^{[note 3][33]}

Critics of the view that innovation causes lasting unemployment argue that technology is used by workers and does not replace them on a large scale.

As the golden age of capitalism came to a close in the 1970s, unemployment once again rose, and this time generally remained relatively high for the rest of the century, across most advanced economies. Several economists once again argued that this may be due to innovation, with perhaps the most prominent being Paul Samuelson.^[34] Overall, the closing decades of the 20th century saw most concern expressed over technological unemployment in Europe, though there were several examples in the U.S.^[35] A number of popular works warning of technological unemployment were also published. These included James S. Albus's 1976 book titled *Peoples' Capitalism: The Economics of the Robot Revolution*;^{[36][37]} David F. Noble with works published in 1984^[38] and 1993;^[39] Jeremy Rifkin and his 1995 book *The End of Work*;^[40] and the 1996 book *The Global*

Trap.^[41] Yet for the most part, other than during the periods of intense debate in the 1930s and 60s, the consensus in the 20th century among both professional economists and the general public remained that technology does not cause long-term joblessness.^[42]

21st century

Opinions

The general consensus that innovation does not cause long-term unemployment held strong for the first decade of the 21st century although it continued to be challenged by a number of academic works,^{[44][45]} and by popular works such as Marshall Brain's *Robotic Nation*^[46] and Martin Ford's *The Lights in the Tunnel: Automation, Accelerating Technology and the Economy of the Future*.^[47]

Since the publication of their 2011 book *Race Against the Machine*, MIT professors Andrew McAfee and Erik Brynjolfsson have been prominent among those raising concern about technological unemployment. The two professors remain relatively optimistic, however, stating "the key to winning the race is not to compete *against* machines but to compete *with* machines".^{[48][49][50][51][52][53][54]}

There is a prevailing opinion that we are in an era of technological unemployment – that technology is increasingly making skilled workers obsolete.

—Prof. Mark MacCarthy
(2014)^[43]

Concern about technological unemployment grew in 2013 due in part to a number of studies predicting substantially increased technological unemployment in forthcoming decades and empirical evidence that, in certain sectors, employment is falling worldwide despite rising output, thus discounting globalization and offshoring as the only causes of increasing unemployment.^{[55][26][56]}

In 2013, professor Nick Bloom of Stanford University stated there had recently been a major change of heart concerning technological unemployment among his fellow economists.^[57] In 2014 the *Financial Times* reported that the impact of innovation on jobs has been a dominant theme in recent economic discussion.^[58] According to the academic and former politician Michael Ignatieff writing in 2014, questions concerning the effects of technological change have been "haunting democratic politics everywhere".^[59] Concerns have included evidence showing worldwide falls in employment across sectors such as manufacturing; falls in pay for low and medium skilled workers stretching back several decades even as productivity continues to rise; the increase in often precarious platform mediated employment; and the occurrence of "jobless recoveries" after recent recessions. The 21st century has seen a variety of skilled tasks partially taken over by machines,

including translation, legal research and even low level journalism. Care work, entertainment, and other tasks requiring empathy, previously thought safe from automation, have also begun to be performed by robots.^{[55][26][60][61][62][63]}

Former U.S. Treasury Secretary and Professor of Economics at Harvard University Lawrence Summers stated in 2014 that he no longer believed automation would always create new jobs and that "This isn't some hypothetical future possibility. This is something that's emerging before us right now." Summers noted that already, more labor sectors were losing jobs than creating new ones.^{[note 4][64][65][66][67]} While himself doubtful about technological unemployment, professor Mark MacCarthy stated in the fall of 2014 that it is now the "prevailing opinion" that the era of technological unemployment has arrived.^[43]

At the 2014 Davos meeting, Thomas Friedman reported that the link between technology and unemployment seemed to have been the dominant theme of that year's discussions. A survey at Davos 2014 found that 80% of 147 respondents agreed that technology was driving jobless growth.^[68] At the 2015 Davos, Gillian Tett found that almost all delegates attending a discussion on inequality and technology expected an increase in inequality over the next five years, and gives the reason for this as the technological displacement of jobs.^[69] 2015 saw Martin Ford win the Financial Times and McKinsey Business Book of the Year Award for his *Rise of the Robots: Technology and the Threat of a Jobless Future*, and saw the first world summit on technological unemployment, held in New York. In late 2015, further warnings of potential worsening for technological unemployment came from Andy Haldane, the Bank of England's chief economist, and from Ignazio Visco, the governor of the Bank of Italy.^{[70][71]} In an October 2016 interview, US President Barack Obama said that due to the growth of artificial intelligence, society would be debating "unconditional free money for everyone" within 10 to 20 years.^[72] In 2019, computer scientist and artificial intelligence expert Stuart J. Russell stated that "in the long run nearly all current jobs will go away, so we need fairly radical policy changes to prepare for a very different future economy." In a book he authored, Russell claims that "One rapidly emerging picture is that of an economy where far fewer people work because work is unnecessary." However, he predicted that employment in healthcare, home care, and construction would increase.^[73]

Other economists have argued that long-term technological unemployment is unlikely. In 2014, Pew Research canvassed 1,896 technology professionals and economists and found a split of opinion: 48% of respondents believed that new technologies would displace more jobs than they would create by the year 2025, while 52% maintained that they would not.^[74] Economics professor Bruce Chapman from Australian National University has advised that studies such as Frey and Osborne's tend to overstate the probability of future job losses, as they don't account for new employment likely to be created, due to technology, in what are currently unknown areas.^[75] Looking deeper into this, small and mid-sized businesses have created a large amount of new jobs around the world, which allows for entrepreneurs and investors to have the freedom to create and grow businesses, which is extremely vital with new technologies emerging everyday.^[76] With all of

these new businesses there will be a large number of workers that will be required to work for these companies, which would improve the world's employment situation, replacing jobs that were previously lost.

General public surveys have often found an expectation that automation would impact jobs widely, but not the jobs held by those particular people surveyed.^[77]

In the early 2020s, generative AI and automation reignited debate over systemic job displacement versus emergent roles in AI-enabled services and new digital industries. Economic analyses point to increased worker transitions into AI supervision and creative task domains.^[78] In July 2025, Ford CEO Jim Farley predicted that "artificial intelligence is going to replace literally half of all white-collar workers in the U.S."^[79] In October 2025, U.S. Senator Bernie Sanders raised concerns about job displacement due to artificial intelligence, citing a report that estimated potential job losses of up to 100 million over the next decade.^[80] He proposed a "robot tax" that would protect workers from the impacts of artificial intelligence.^[81]

Studies

A number of studies have predicted that automation will take a large proportion of jobs in the future, but estimates of the level of unemployment this will cause vary. Research by Carl Benedikt Frey and Michael Osborne of the Oxford Martin School showed that employees engaged in "tasks following well-defined procedures that can easily be performed by sophisticated algorithms" are at risk of displacement. The study, published in 2013, shows that automation can affect both skilled and unskilled work and both high and low-paying occupations; however, low-paid physical occupations are most at risk. It estimated that 47% of US jobs were at high risk of automation.^[26] In 2014, the economic think tank Bruegel released a study, based on the Frey and Osborne approach, claiming that across the European Union's 28 member states, 54% of jobs were at risk of automation. The countries where jobs were least vulnerable to automation were Sweden, with 46.69% of jobs vulnerable, the UK at 47.17%, the Netherlands at 49.50%, and France and Denmark, both at 49.54%. The countries where jobs were found to be most vulnerable were Romania at 61.93%, Portugal at 58.94%, Croatia at 57.9%, and Bulgaria at 56.56%.^{[82][83]} A 2015 report by the Taub Center found that 41% of jobs in Israel were at risk of being automated within the next two decades.^[84] In January 2016, a joint study by the Oxford Martin School and Citibank, based on previous studies on automation and data from the World Bank, found that the risk of automation in developing countries was much higher than in developed countries. It found that 77% of jobs in China, 69% of jobs in India, 85% of jobs in Ethiopia, and 55% of jobs in Uzbekistan were at risk of automation.^[85] The World Bank similarly employed the methodology of Frey and Osborne. A 2016 study by the International Labour Organization found 74% of salaried electrical & electronics industry positions in Thailand, 75% of salaried electrical & electronics industry positions in Vietnam, 63% of salaried electrical & electronics industry positions in Indonesia, and 81% of salaried electrical & electronics industry positions in the Philippines were at high risk of

automation.^[86] A 2016 United Nations report stated that 75% of jobs in the developing world were at risk of automation, and predicted that more jobs might be lost when corporations stop outsourcing to developing countries after automation in industrialized countries makes it less lucrative to outsource to countries with lower labor costs.^[87]

The Council of Economic Advisers, a US government agency tasked with providing economic research for the White House, in the 2016 Economic Report of the President, used the data from the Frey and Osborne study to estimate that 83% of jobs with an hourly wage below \$20, 31% of jobs with an hourly wage between \$20 and \$40, and 4% of jobs with an hourly wage above \$40 were at risk of automation.^[88] A 2016 study by Ryerson University (now Toronto Metropolitan University) found that 42% of jobs in Canada were at risk of automation, dividing them into two categories - "high risk" jobs and "low risk" jobs. High risk jobs were mainly lower-income jobs that required lower education levels than average. Low risk jobs were on average more skilled positions. The report found a 70% chance that high risk jobs and a 30% chance that low risk jobs would be affected by automation in the next 10–20 years.^[89] A 2017 study by PricewaterhouseCoopers found that up to 38% of jobs in the US, 35% of jobs in Germany, 30% of jobs in the UK, and 21% of jobs in Japan were at high risk of being automated by the early 2030s.^[90] A 2017 study by Ball State University found about half of American jobs were at risk of automation, many of them low-income jobs.^[91] A September 2017 report by McKinsey & Company found that as of 2015, 478 billion out of 749 billion working hours per year dedicated to manufacturing, or \$2.7 trillion out of \$5.1 trillion in labor, were already automatable. In low-skill areas, 82% of labor in apparel goods, 80% of agriculture processing, 76% of food manufacturing, and 60% of beverage manufacturing were subject to automation. In mid-skill areas, 72% of basic materials production and 70% of furniture manufacturing was automatable. In high-skill areas, 52% of aerospace and defense labor and 50% of advanced electronics labor could be automated.^[92] In October 2017, a survey of information technology decision makers in the US and UK found that a majority believed that most business processes could be automated by 2022. On average, they said that 59% of business processes were subject to automation.^[93] A November 2017 report by the McKinsey Global Institute that analyzed around 800 occupations in 46 countries estimated that between 400 million and 800 million jobs could be lost due to robotic automation by 2030. It estimated that jobs were more at risk in developed countries than developing countries due to a greater availability of capital to invest in automation.^[94] Job losses and downward mobility blamed on automation has been cited as one of many factors in the resurgence of nationalist and protectionist politics in the US, UK and France, among other countries.^{[95][96][97][98][99]}

However, not all recent empirical studies have found evidence to support the idea that automation will cause widespread unemployment. A study released in 2015, examining the impact of industrial robots in 17 countries between 1993 and 2007, found no overall reduction in employment was caused by the robots, and that there was a slight increase in overall wages.^[100] According to a study published in McKinsey Quarterly^[101] in 2015 the impact of computerization in most cases is

not replacement of employees but automation of portions of the tasks they perform.^[102] A 2016 OECD study found that among the 21 OECD countries surveyed, on average only 9% of jobs were in foreseeable danger of automation, but this varied greatly among countries: for example in South Korea the figure of at-risk jobs was 6% while in Austria it was 12%.^[103] In contrast to other studies, the OECD study does not primarily base its assessment on the tasks that a job entails, but also includes demographic variables, including sex, education and age. In 2017, Forrester estimated that automation would result in a net loss of about 7% of jobs in the US by 2027, replacing 17% of jobs while creating new jobs equivalent to 10% of the workforce.^[104] Another study argued that the risk of US jobs to automation had been overestimated due to factors such as the heterogeneity of tasks within occupations and the adaptability of jobs being neglected. The study found that once this was taken into account, the number of occupations at risk to automation in the US drops, ceteris paribus, from 38% to 9%.^[105] A 2017 study on the effect of automation on Germany found no evidence that automation caused total job losses but that they do affect the jobs people are employed in; losses in the industrial sector due to automation were offset by gains in the service sector. Manufacturing workers were also not at risk from automation and were in fact more likely to remain employed, though not necessarily doing the same tasks. However, automation did result in a decrease in labour's income share as it raised productivity but not wages.^[106]

A 2018 Brookings Institution study that analyzed 28 industries in 18 OECD countries from 1970 to 2018 found that automation was responsible for holding down wages. Although it concluded that automation did not reduce the overall number of jobs available and even increased them, it found that from the 1970s to the 2010s, it had reduced the share of human labor in the value added to the work, and thus had helped to slow wage growth.^[107] In April 2018, Adair Turner, former Chairman of the Financial Services Authority and head of the Institute for New Economic Thinking, stated that it would already be possible to automate 50% of jobs with current technology, and that it will be possible to automate all jobs by 2060.^[108]

Premature deindustrialization

Premature deindustrialization occurs when developing nations deindustrialize without first becoming rich, as happened with the advanced economies. The concept was popularized by Dani Rodrik in 2013, who went on to publish several papers showing the growing empirical evidence for the phenomena. Premature deindustrialization adds to concern over technological unemployment for developing countries – as traditional compensation effects that advanced economy workers enjoyed, such being able to get well paid work in the service sector after losing their factory jobs – may not be available.^{[109][110]}

Some commentators, such as Carl Benedikt Frey, argue that with the right responses, the negative effects of further automation on workers in developing economies can still be avoided.^[111]